

Branch BR8 to Data Center DC1 - degradation and transport failure scenarios

1. Executive summary

Path decision after
loss of last DC1 transport

Lost on UDP 6200
(4.3 %, ≈ 9.1 s @ 50 pps)

Loss during +50 ms latency
and MPLS outage

Automatic failback
after restoration

Key findings. A +50 ms latency on DC1's Internet overlay was absorbed without any path change or loss, which indicates that the active SLA policy tolerates this degradation. Losing DC1's MPLS had no effect because traffic was already carried over the Internet overlay. When the last direct DC1 transport (Internet) was lost, the SD-WAN made a new path decision within about one to two seconds and rerouted traffic to DC1 through the DC2 hub and the DC1-DC2 interconnect (DCI). The continuous probe recorded 453 lost packets, equivalent to roughly 9 seconds of interruption, before steady state resumed. After the transports were restored, traffic returned automatically to its nominal path in under a minute.

Figure 1 - Nominal path (green) and final failover path via DC2 and the DCI (orange).

Element	Identifier	Role in the test
Source node	BR8-Ubuntu - 192.168.219.1	Probe initiator; SD-WAN site BR8 with two Internet circuits (INET1, INET2)
Target node	DC1 - 192.168.203.100	Probe echo responder and VoIP target
Underlay router	vyosrouter	Emulates carrier networks; used to inject faults
DC1 Internet overlay	vyosrouter eth10 (DC1-INET-221)	Latency injection, then shutdown
DC1 MPLS	vyosrouter eth11 (DC1-MPLS-191)	Shutdown
Observation	Prisma SD-WAN Flow Browser; Stigix test metrics	Path decisions per flow; loss, latency, jitter

Initial condition noted. Before the test, vyosrouter eth0 (ALL-MPLS-190) already carried a pre-existing 250 ms delay policy unrelated to this test. It was left untouched and may explain why the MPLS transport was not the preferred path at baseline.

3. Methodology

The scenario was executed step by step with approximately 30 seconds between each action to let the fabric stabilise and to capture a clean observation for each state. Paths were read from the Prisma SD-WAN Flow Browser, and loss and latency from the Stigix convergence test.

Step	Action	Purpose
0	Baseline capture of active paths	Establish nominal routing BR8 to DC1
1	Start continuous UDP 6200 probe, 50 pps	Measure loss and latency throughout
2	Add 50 ms latency on DC1 Internet (eth10)	Test SLA-driven path steering on degradation
3	Shut DC1 MPLS (eth11)	Remove one of the two DC1 transports
4	Shut DC1 Internet (eth10)	Remove all direct DC1 transports
5	Stop probe, collect final metrics	Quantify impact
6	Clear latency, restore eth11 and eth10	Return to nominal and observe failback

4. Timeline of events (UTC)

Time	Event	Observed path BR8 to DC1	Probe status
18:14:07	Baseline	BR8-INET1 -> DC1-INET	-
18:14:23	Convergence probe started	BR8-INET1 -> DC1-INET	Running, 0 % loss
≈18:15:25	+50 ms latency on DC1-INET	BR8-INET1 -> DC1-INET (unchanged)	0 % loss; avg latency 22.5 -> 30.7 ms
≈18:16:27	DC1-MPLS shut	BR8-INET1 -> DC1-INET (unchanged)	0 % loss; avg latency 38.9 ms
≈18:17:20	DC1-INET shut	Decision in progress	-
18:17:21.64	SD-WAN path decision #1	BR8-INET2 -> DC1-INET (transient, 240 ms)	Loss occurring
18:17:21.88	SD-WAN path decision #2	BR8-INET1 -> DC2-INET (via DCI)	Recovered
≈18:17:55	Probe stopped	BR8-INET1 -> DC2-INET	Final: 4.3 % loss
18:18:05-25	Latency cleared, eth11 and eth10 restored	-	-
18:19:05	Failback confirmed	BR8-INET1 -> DC1-INET	-

Times marked ≈ are command execution times from the orchestration log; the two decision timestamps are taken from the SD-WAN Flow Browser and are exact.

5. Results

5.1 Continuous probe (UDP 6200)

Metric	Value
Packets sent	10,485
Packets received	10,032
Packets lost	453 (4.3 %)
Equivalent interruption at 50 pps	≈ 9.1 s
Average latency over the test	36.5 ms
Average jitter	7.2 ms

Intermediate readings showed 0 % loss up to and including the MPLS outage, so the entire loss is attributed to the final failover, when DC1 lost its last direct transport. The steady rise of the running average latency after step 2 confirms that the 50 ms impairment was effective on the active path.

5.2 Voice simulation (UDP 6100)

Voice session to DC1	Path	Packets received
Before latency injection	BR8-INET1 -> DC1-INET	999 / 1000
Spanning the latency injection	BR8-INET1 -> DC1-INET	998 / 1000
After the failover (new call)	BR8-INET1 -> DC2-INET	1000 / 1000

Voice sessions were not materially affected. The call placed after the failover was set up directly on the DC2 path without loss, which shows that new sessions benefit immediately from the backup path.

6. Findings and analysis

F1 - Degradation tolerance. A 50 ms increase on the DC1 Internet overlay did not trigger path steering. This is consistent with an SLA threshold above the resulting latency, or with the lack of a better alternative, since MPLS carried an additional 250 ms delay.

F2 - Single transport loss is transparent. Losing DC1 MPLS had no impact because it was not the active path. The fabric kept a healthy Internet path to DC1.

F3 - Fast control-plane decision. After the loss of DC1 Internet, the SD-WAN issued a new decision in about one to two seconds. It first chose BR8-INET2 to DC1-INET, which was no longer viable, then settled on BR8-INET1 to DC2-INET 240 ms later.

F4 - End-to-end recovery longer than path decision. The probe measured about 9 seconds of loss, well above the path decision time. The likely contributors are path liveness detection on the remaining DC1 tunnels and return-path convergence: traffic from DC1 back to BR8 must also be rerouted through the DC1 and DC2, which depends on data-center LAN routing convergence. This should be confirmed with per-direction measurements.

F5 - Hub redundancy validated. DC1 remained reachable when it had no WAN transport at all, thanks to the DC2 hub and the DCI.

F6 - Clean fallback. Traffic returned to its nominal path automatically within a minute of restoration, without manual action.

7. Recommendations

R1. Review the SLA policy for latency-sensitive applications. If a +50 ms degradation should trigger steering, lower the latency threshold or confirm that it is set as intended.

R2. Investigate the ≈ 9 s recovery. Measure each direction separately, review tunnel liveness and probe timers on the DC1 ION devices, and check the routing convergence time between DC1 and DC2 across the DCI (for example BFD on the DC LAN routing protocol).

R3. Re-run the scenario with the pre-existing 250 ms MPLS delay removed to validate a fully nominal MPLS fallback path.

R4. Add a TCP application to the test set (custom TCP workload) to verify session survival during the DC2 failover.

8. Restoration and lab state

All changes made for this test were reverted and verified against the router's live state. eth10 and eth11 are administratively up with no QoS policy. The pre-existing eth0 delay policy was not modified. No firewall blocks were added.

Action	VyOS CLI equivalent	Status
Latency eth10	set traffic-policy network-emulator LAB_COMBINED_eth10 network-delay 50	Removed
Shut eth11	set interfaces ethernet eth11 disable	Reverted
Shut eth10	set interfaces ethernet eth10 disable	Reverted